

Question description

You have an undirected graph consisting of n vertices with weighted edges.

A simple cycle is a cycle of the graph without repeated vertices. Let the *weight* of the cycle be the **XOR** of weights of edges it consists of.

Let's say the graph is *good* if all its *simple* cycles have weight 1. A graph is bad if it's not good.

Initially, the graph is empty. Then q queries follow. Each query has the next type:

- $u \ v \ x$ — add edge between vertices u and v of weight x if it doesn't make the graph bad.

For each query print, was the edge added or not.

Constraints

$$3 \leq n \leq 3 \cdot 10^5; 1 \leq q \leq 5 \cdot 10^5$$

$$1 \leq u, v \leq n; u \neq v; 0 \leq x \leq 1$$

Input

The first line contains two integers n and q — the number of vertices and queries.

Next q lines contain queries — one per line. Each query contains three integers u , v and x — the vertices of the edge and its weight.

It's guaranteed that there are no multiple edges in the input.

Output

For each query, print YES if the edge was added to the graph, or NO otherwise (both case-insensitive).

Logical Test Cases

Test Case 1

INPUT (STDIN)

9 12
6 1 0
1 3 1
3 6 0
6 2 0
6 4 1
3 4 1
2 4 0
2 5 0
4 5 0
7 8 1
8 9 1
9 7 0

EXPECTED OUTPUT

YES
YES
YES
YES
YES
NO
YES
YES
NO
YES
YES
NO

Test Case 2

INPUT (STDIN)

9 6
1 3 1
3 6 0
6 4 1
2 4 0
4 5 0
7 8 1

EXPECTED OUTPUT

YES
YES
YES
YES
YES
YES

Mandatory Test Cases

Test Case 1

KEYWORD

int dfs1(int np,int lst)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

12

Test Case 2

TOKEN COUNT

1255

Test Case 3

NLOC

78